Progamming Engineering

Course 8

14 April

Revision

- Design Patterns: Definitions, Elements
- GOF: Creational Patterns, Structural Patterns, Behavioral Patterns

Creational Patterns

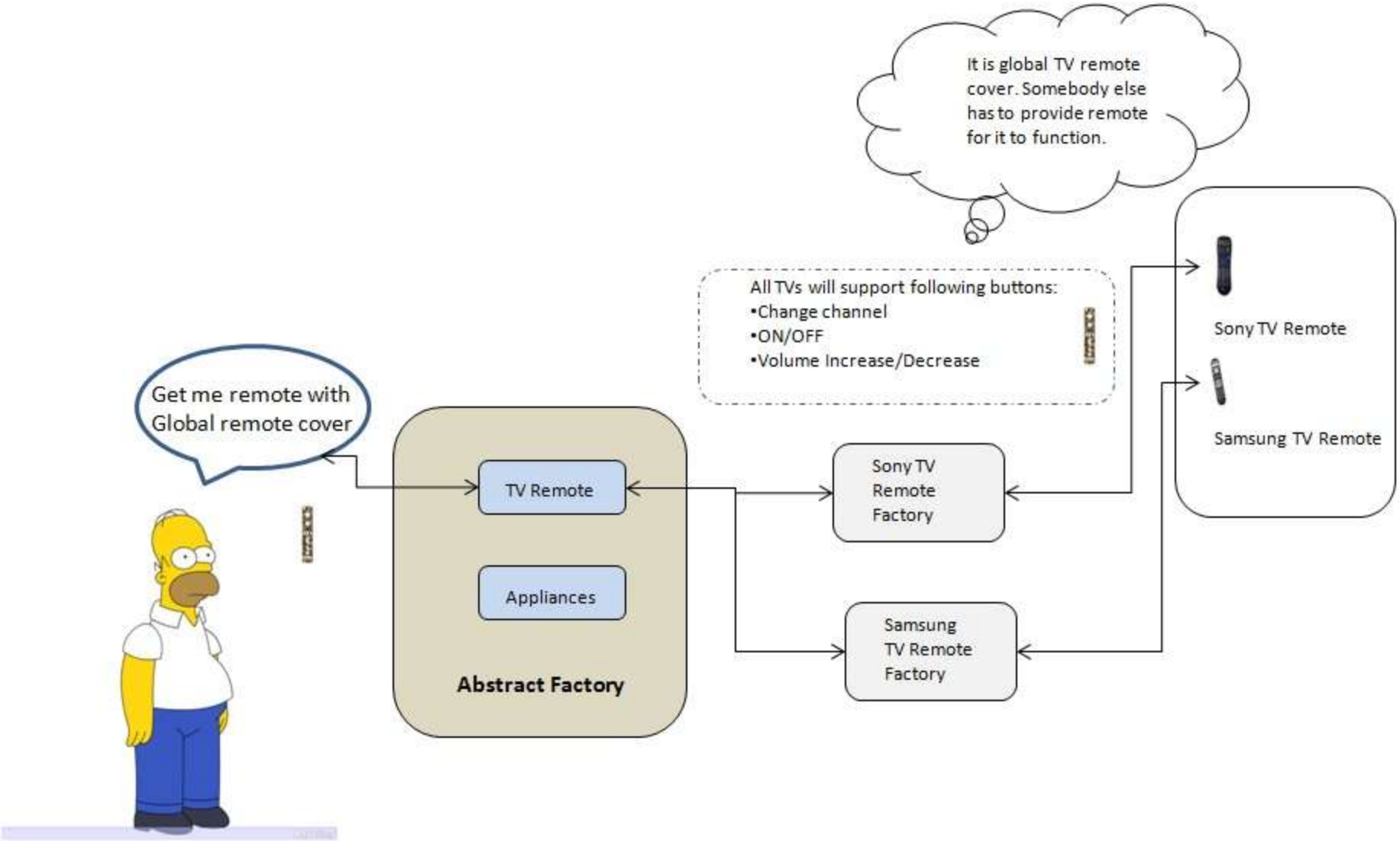
- Abstract Factory
- Builder
- Factory Method
- Prototype
- Singleton
- Lazy initialization
- Object pool
- Multiton
- Resource acquisition

Creational Patterns – Introduction

- Abstract the instantiation process
- Help **make a system independent of how its objects are created, composed, and represented**
- There are two recurring themes in these patterns:
  - First, they all encapsulate knowledge about which concrete classes the system uses
  - Second, they hide how instances of these classes are created and put together
- The creational patterns give a lot of flexibility in *what* gets created, *who creates it, how it gets created, and when*

Creational Patterns – Abstract Factory

- **Intent** – Provide an interface for creating families of related or dependent objects without specifying their concrete classes
- **Also Known As** – Kit
- **Motivation**
  - Consider a user interface toolkit that supports multiple look-and-feel standards, defines different appearances and behaviors for user interface "widgets" like scroll bars, windows, and buttons
  - To be portable across look-and-feel standards, an application should not hard-code its widgets for a particular look and feel

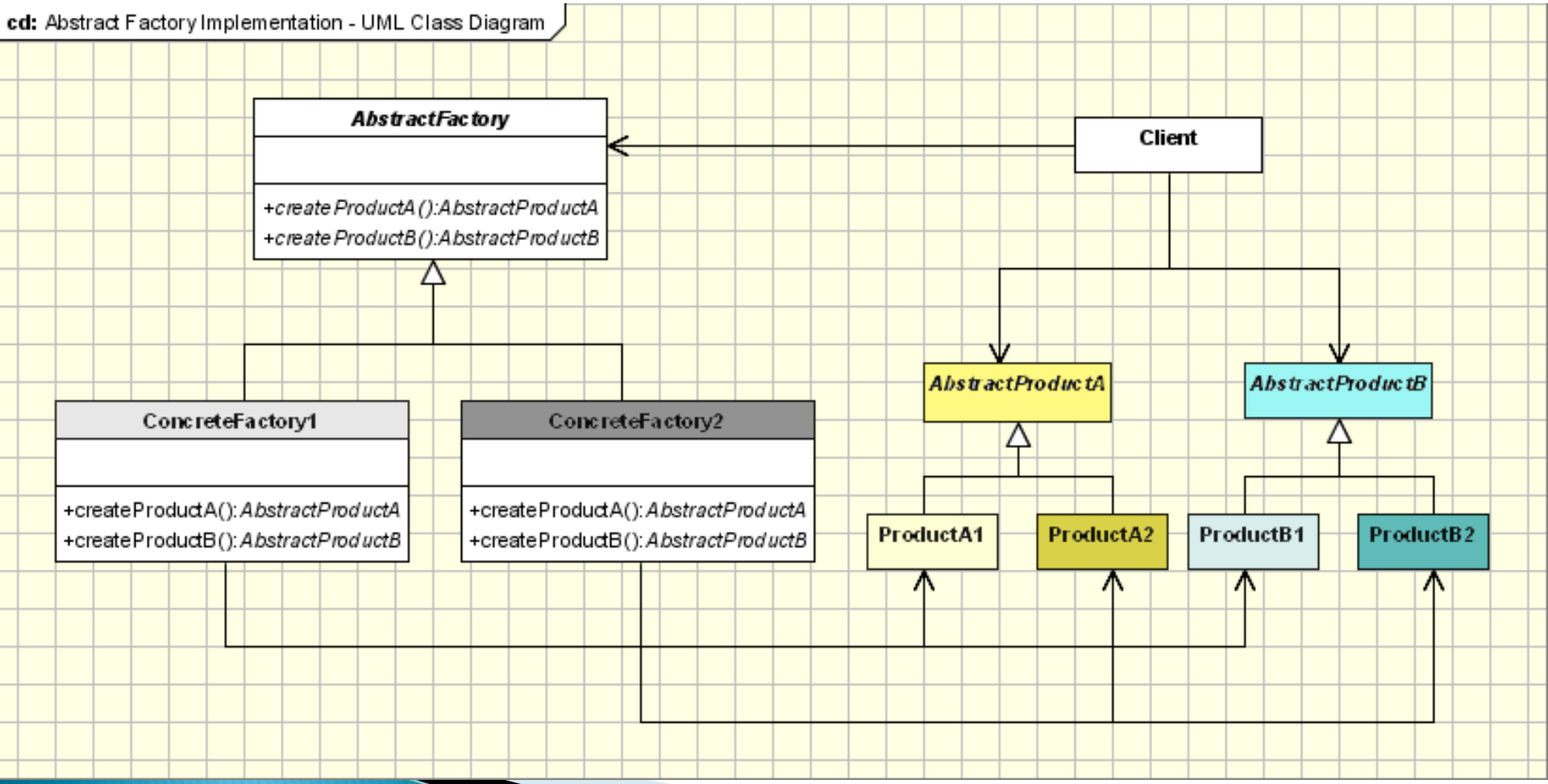

It is global TV remote cover. Somebody else has to provide remote for it to function.
All TVs will support following buttons:
•Change channel
•ON/OFF
•Volume Increase/Decrease
Sony TV Remote
Samsung TV Remote
Get me remote with Global remote cover
TV Remote
Appliances
Abstract Factory
Sony TV Remote Factory
Samsung TV Remote Factory

Abstract Factory

- **Applicability** – Use this pattern when:
  - a system should be independent of how its products are created, composed, and represented
  - a system should be configured with one of multiple families of products
  - a family of related product objects is designed to be used together, and you need to enforce this constraint
  - you want to provide a class library of products, and you want to reveal just their interfaces, not their implementations

CP – Abstract Factory

- **Abstract Factory** – maze, computer components



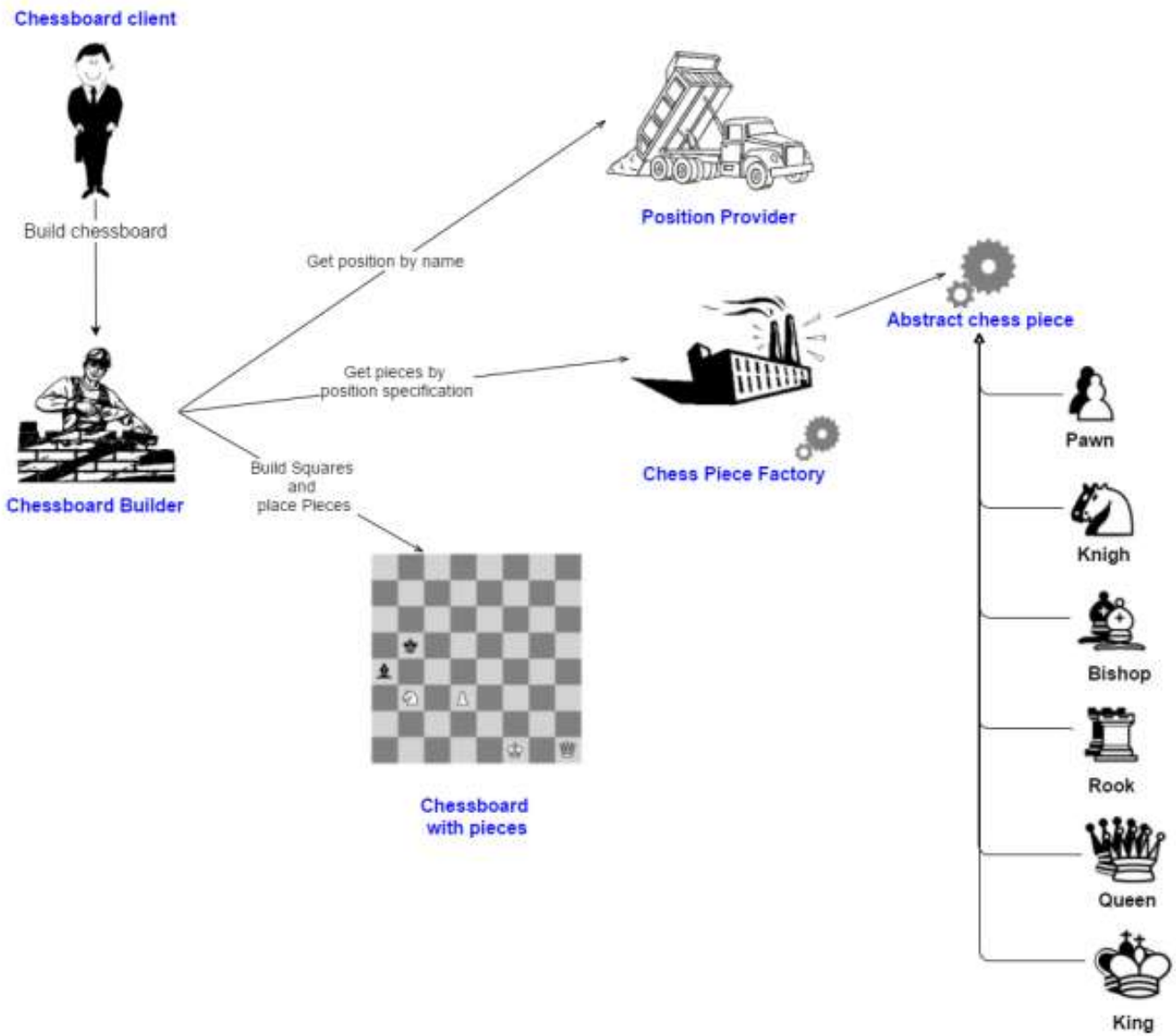
Abstract Factory– The Good, The Bad …

- Detaches creators from created objects
- Guarantees compatibility of created objects
- Preserves S and O (from SOLID)
- Code becomes complicated because of lots of extra classes

Creational Patterns – Builder

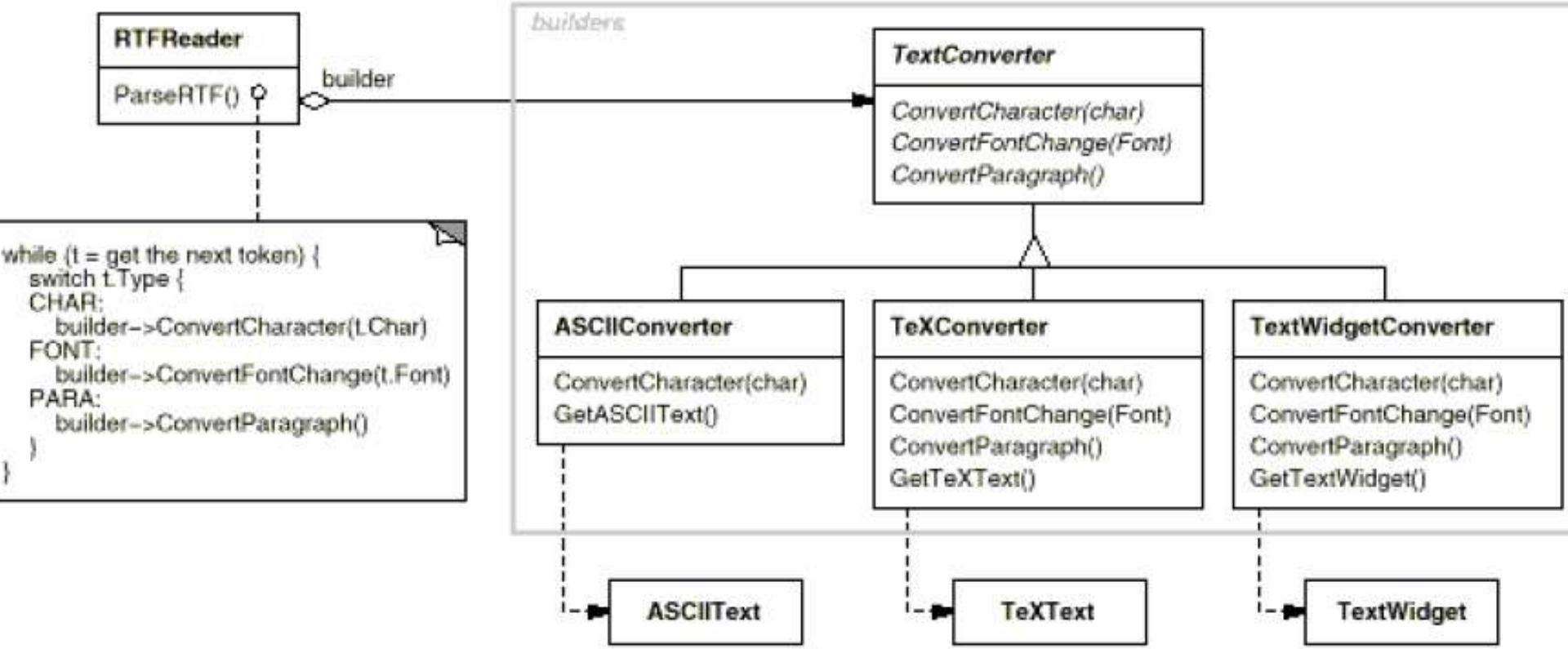
- **Intent – Separate the construction of a complex object from its representation** so that the same construction process can create different representations
- **Motivation**
  - A reader for the RTF (Rich Text Format) document exchange format should be able to convert RTF to many text formats.
  - The reader might convert RTF documents into plain ASCII text or into a text widget that can be edited interactively. The problem, however, is that the number of possible conversions is open-ended. So it should be easy to add a new conversion without modifying the reader.

Builder and Factory Design Pattern - Building chessboard with pieces



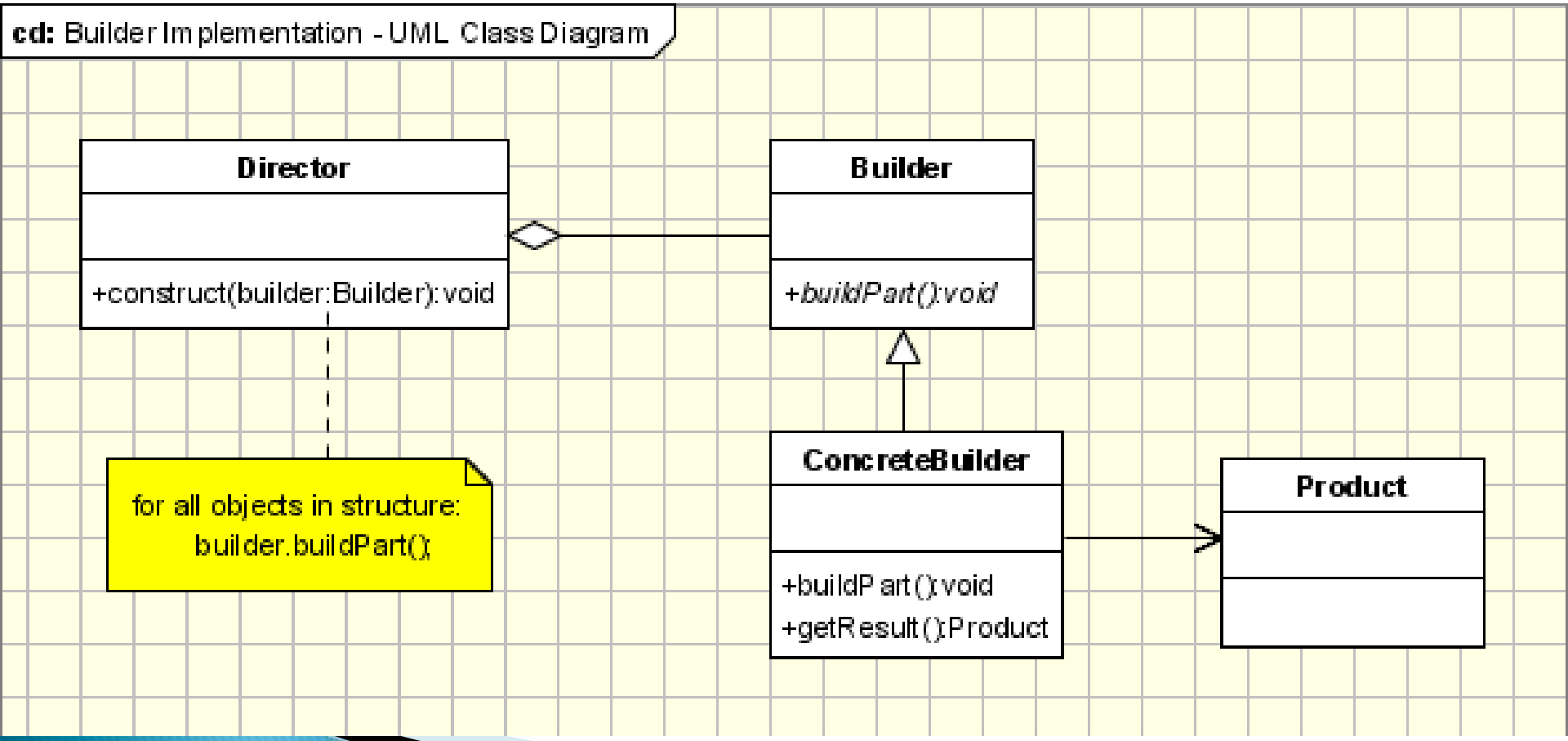
Builder 2

- Each kind of converter class takes the mechanism for creating and assembling
- The converter is separate from the reader



CP – Builder

- **Builder** – RTF Reader, Happy Meal



Builder 3

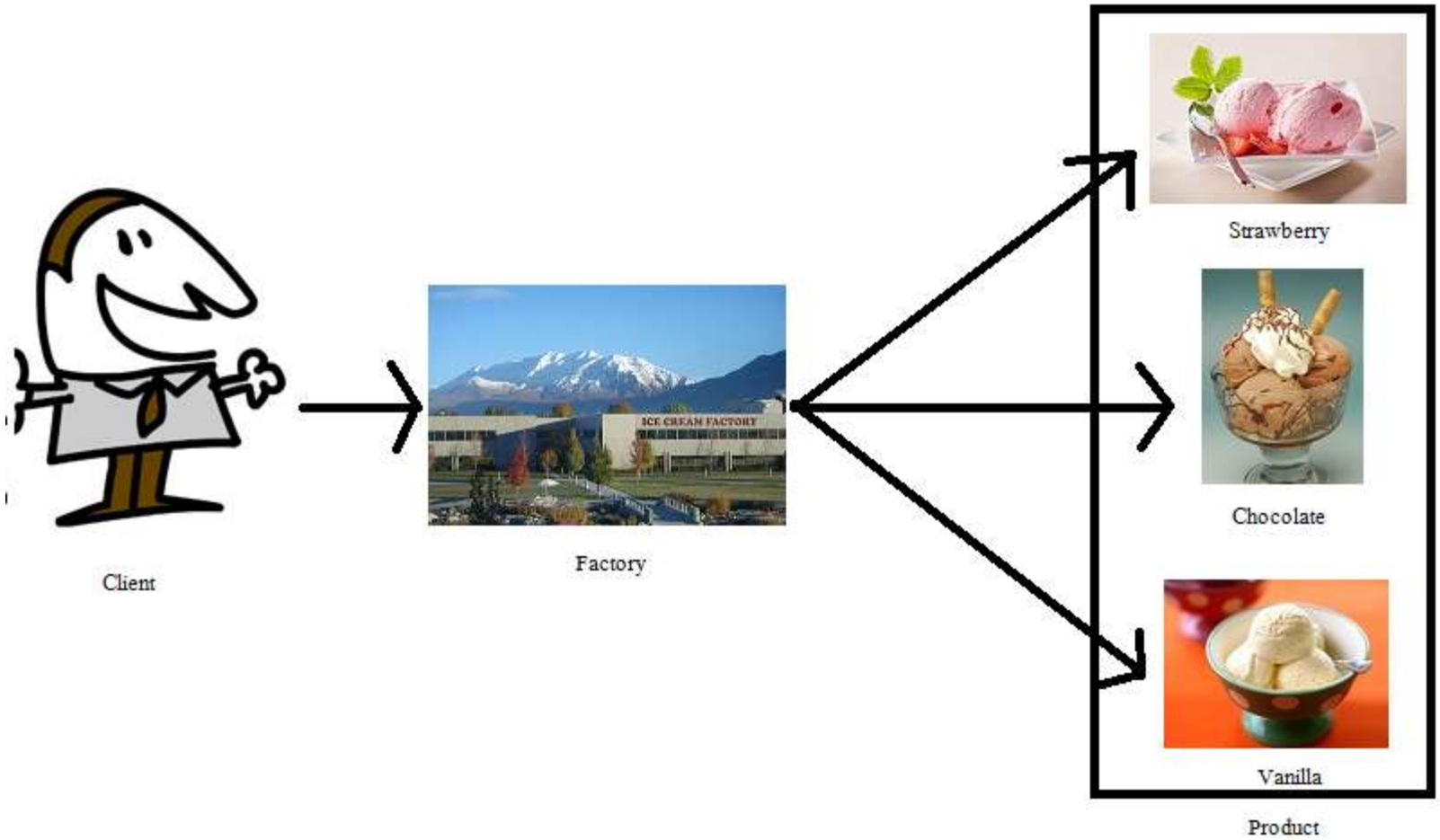
- **Applicability** – Use when the parameters required for construction are numerous, but only some are required for each specific object (different kinds of pizza)
- When you want to build different representations of similar objects (rail bridge and road bridge)

Builder – The Good, The Bad …

- Allows for step by step construction of objects
- Steps can be skipped and run later, or can be run recursively
- Allows for reusing code when building similar (but different) objects
- Preserves S (from SOLID) – decouples complex building processes from the business logic
- Code becomes complicated because of many new classes

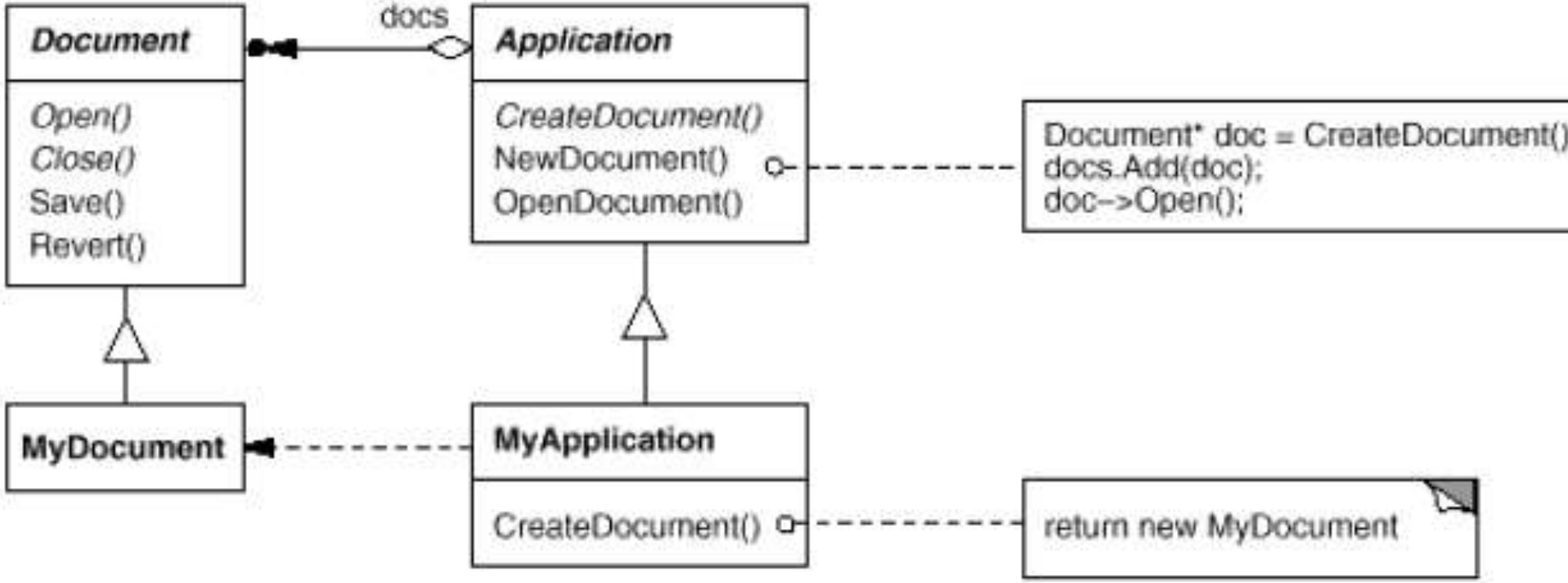
Creational Patterns – Factory Method

- **Intent** – Define an interface for creating an object, but let subclasses decide which class to instantiate
- **Also Known As** – Virtual Constructor
- **Motivation** – To create a drawing application, for example, we define the classes DrawingApplication and DrawingDocument. The Application class is responsible for managing Documents and will create them as required (at Open or New from a menu, for example)


Client
ICE CREAM FACTORY
Factory
Strawberry
Chocolate
Vanilla
Product

Factory Method 2

- Subclasses redefine an abstract CreateDocument
- It can then instantiate application-specific Documents without knowing their class
- CreateDocument is a **factory method because it's** responsible for "manufacturing" an object



Factory Method 3

- **Applicability** – Use this pattern when
  - a class can't anticipate the class of objects it must create
  - a class wants its subclasses to specify the objects it creates
  - classes delegate responsibility to one of several helper subclasses, and you want to localize the knowledge of which helper subclass is the delegate

Factory Method 4

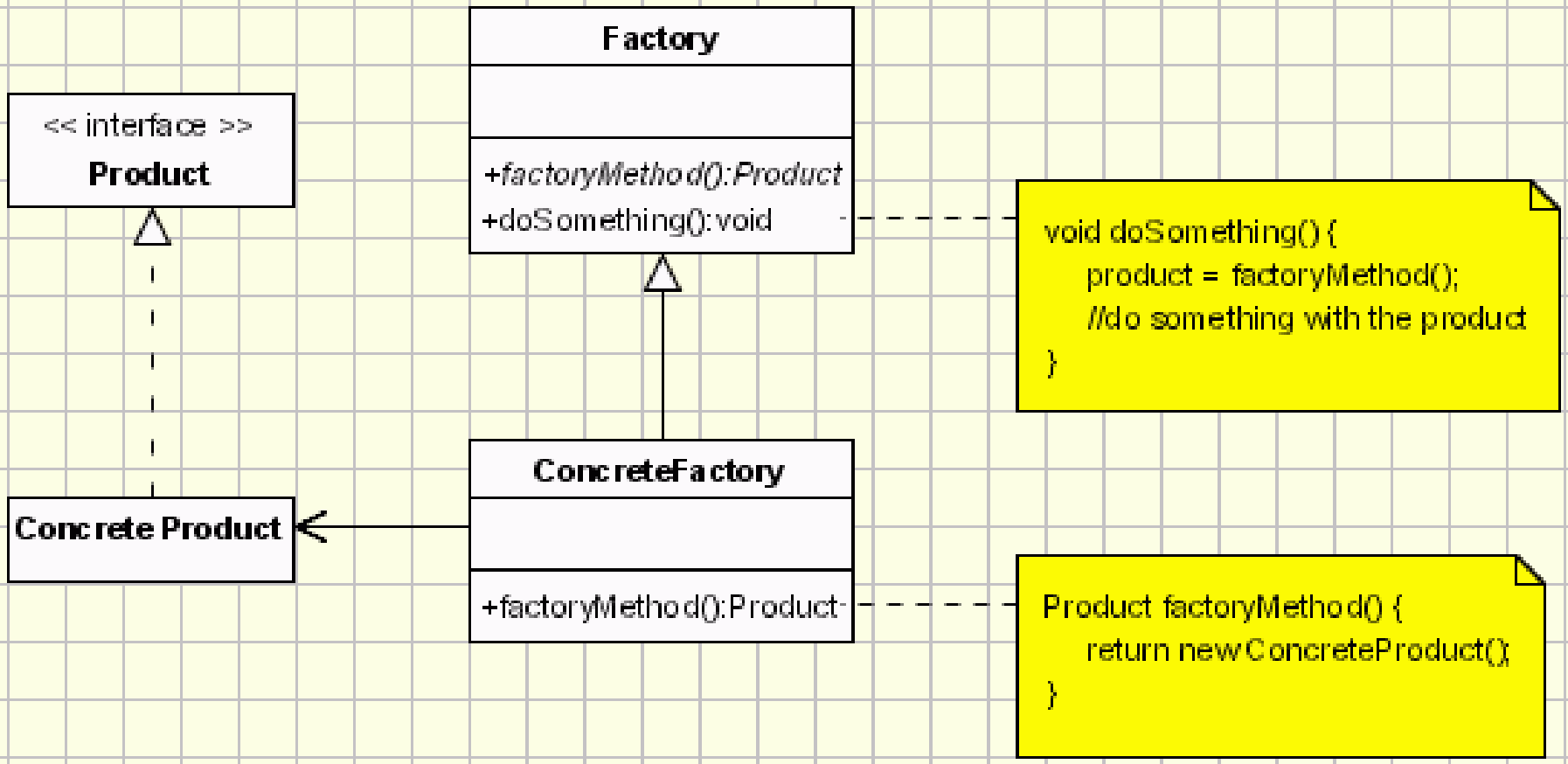
- **Consequences**

1. *Provides interconnections for subclasses*

2. *Connects parallel class hierarchies*

CP – Factory Method

- **Factory Method** – Open/New Project, Hello Mr/Ms

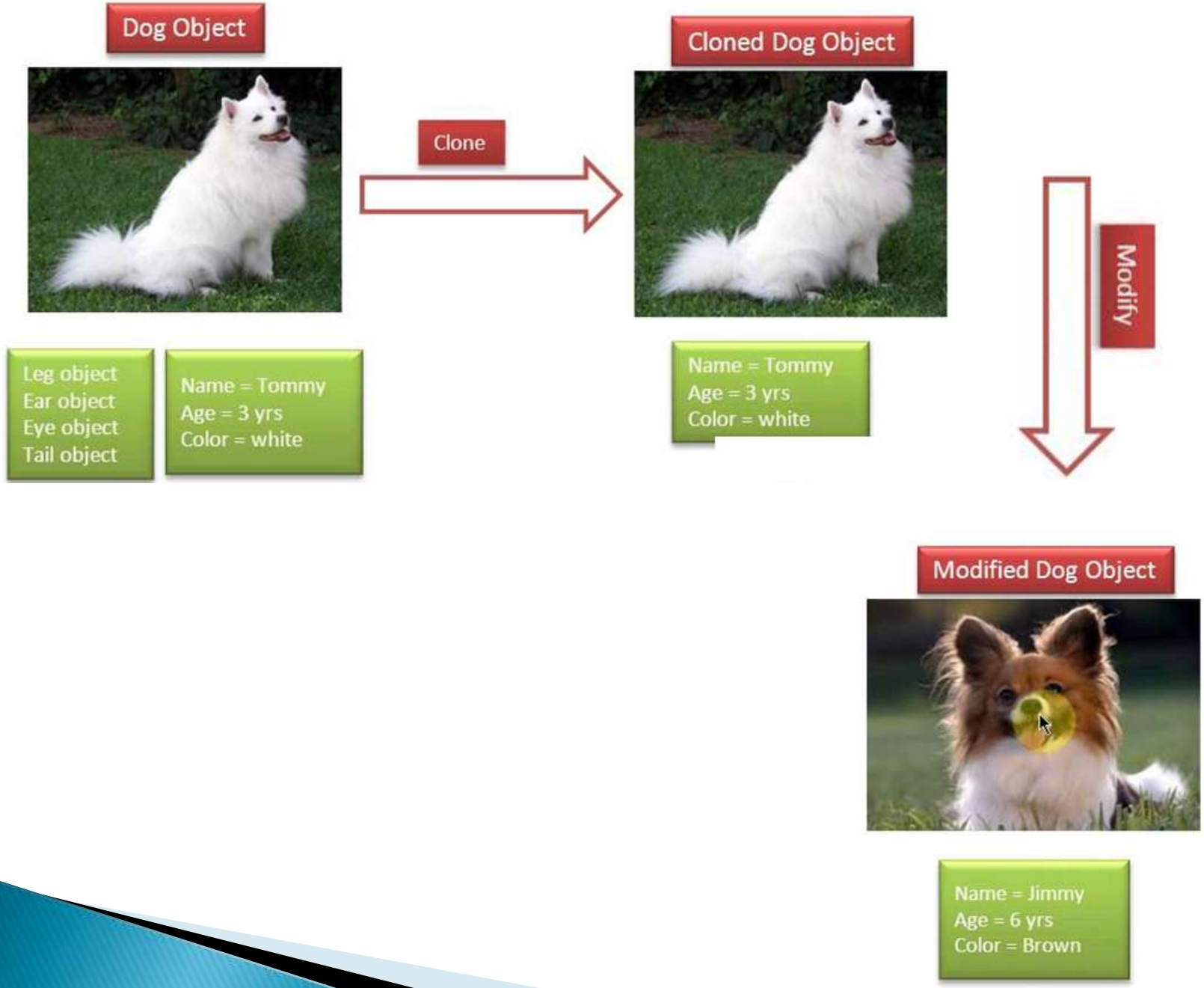


Factory Method– The Good, The Bad …

- Detaches creators from created objects
- Preserves S and O (from SOLID)
- Code becomes complicated because of extra (derived) classes

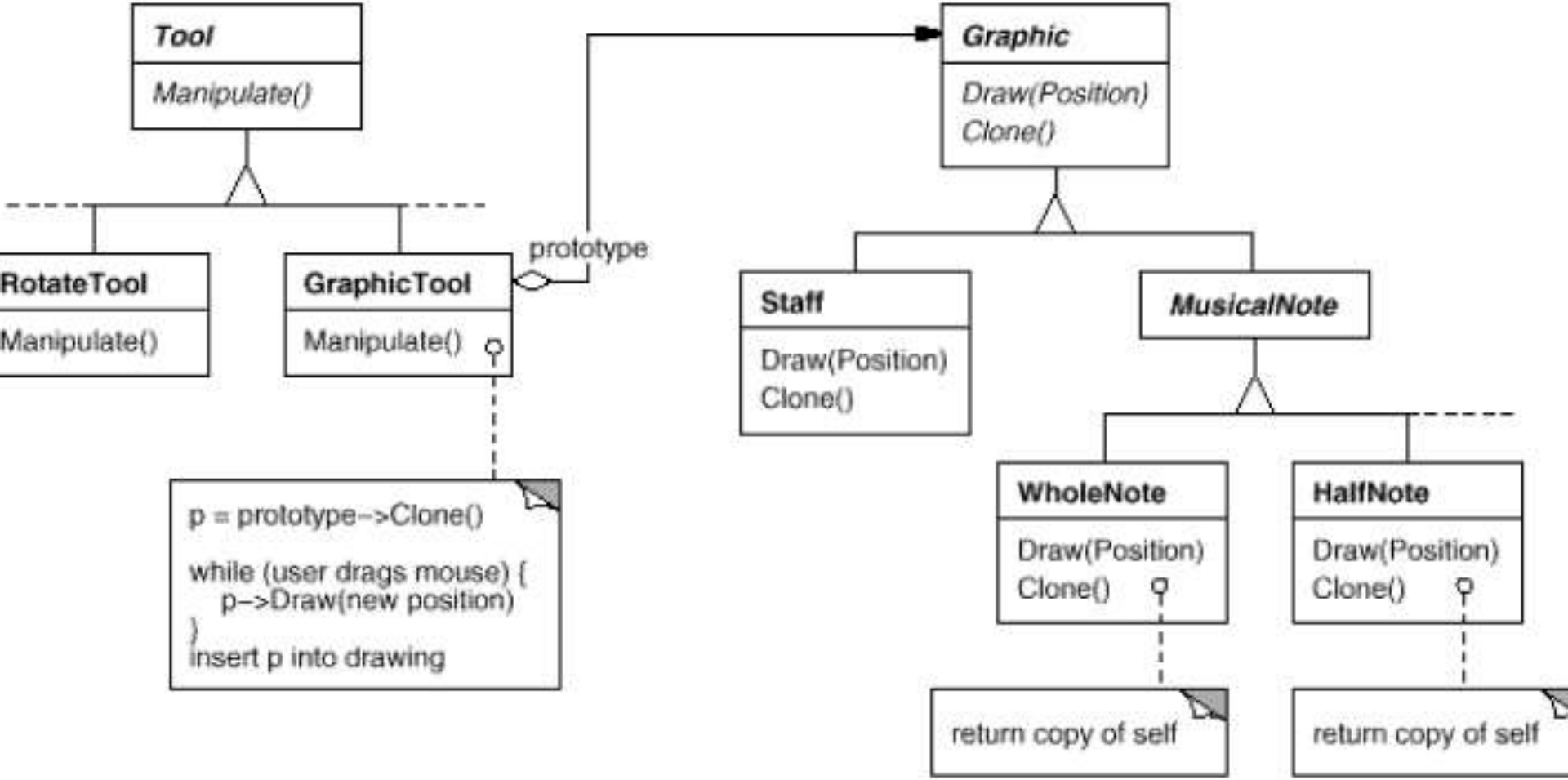
Creational Patterns – Prototype

- **Intent** – Specify the kinds of objects to create using a prototypical instance, and create new objects by copying this prototype
- **Motivation** – You could build an editor for music scores by customizing a general framework for graphical editors and adding new objects that represent notes, rests, and staves
- The editor framework may have a palette of tools for adding these music objects to the score. The palette would also include tools for selecting, moving, and otherwise manipulating music objects


Dog Object
Leg object
Ear object
Eye object
Tail object
Name = Tommy
Age = 3 yrs
Color = white
Clone
Cloned Dog Object
Name = Tommy
Age = 3 yrs
Color = white
Modify
Modified Dog Object
Name = Jimmy
Age = 6 yrs
Color = Brown

Prototype 2

- We can use the Prototype pattern to reduce the number of classes

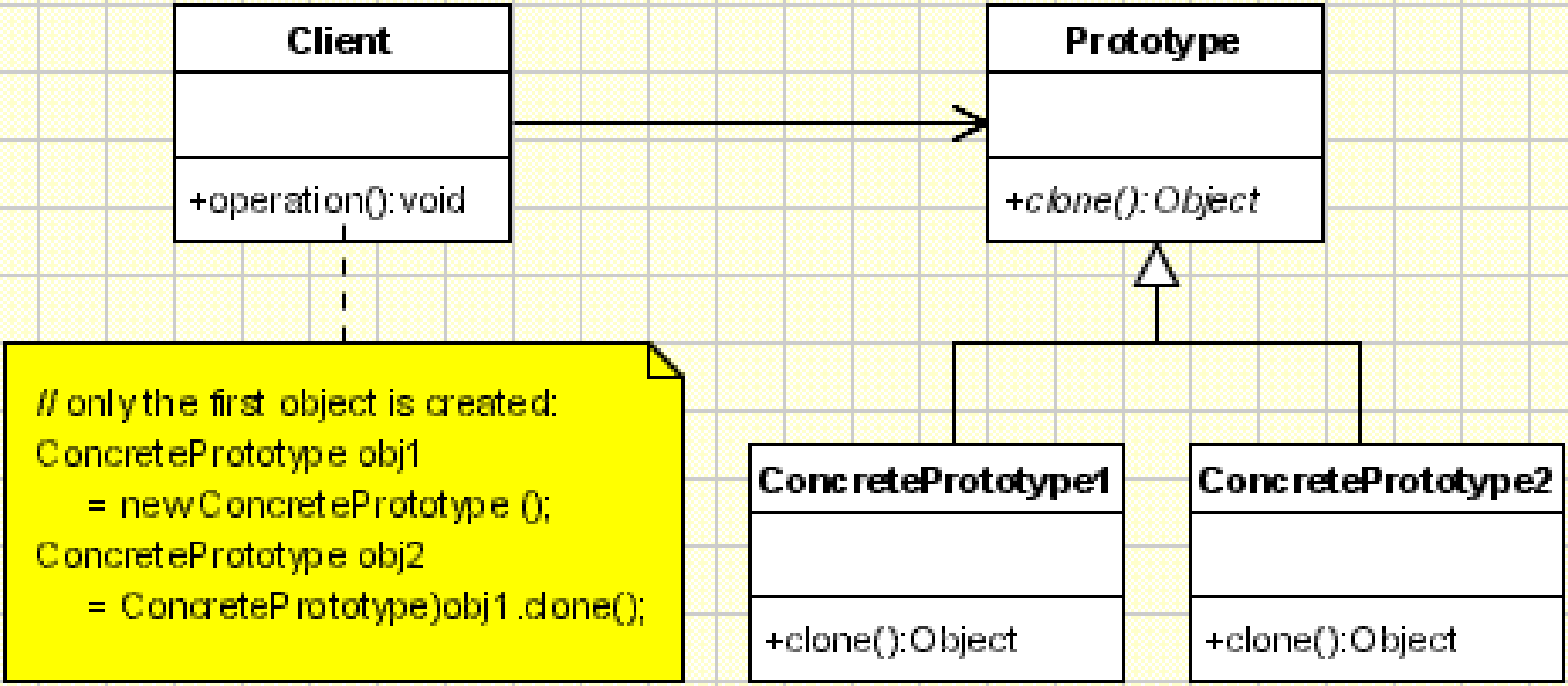


Prototype 3

- **Applicability** – Use the Prototype pattern when a system should be independent of how its products are created, composed, and represented; *and*
  - when the classes to instantiate are specified at run-time, for example, by dynamic loading; *or*
  - to avoid building a class hierarchy of factories that parallels the class hierarchy of products; *or*
  - when instances of a class can have one of only a few different combinations of state

CP – Prototype

- **Prototype** – Music editor, Clonable

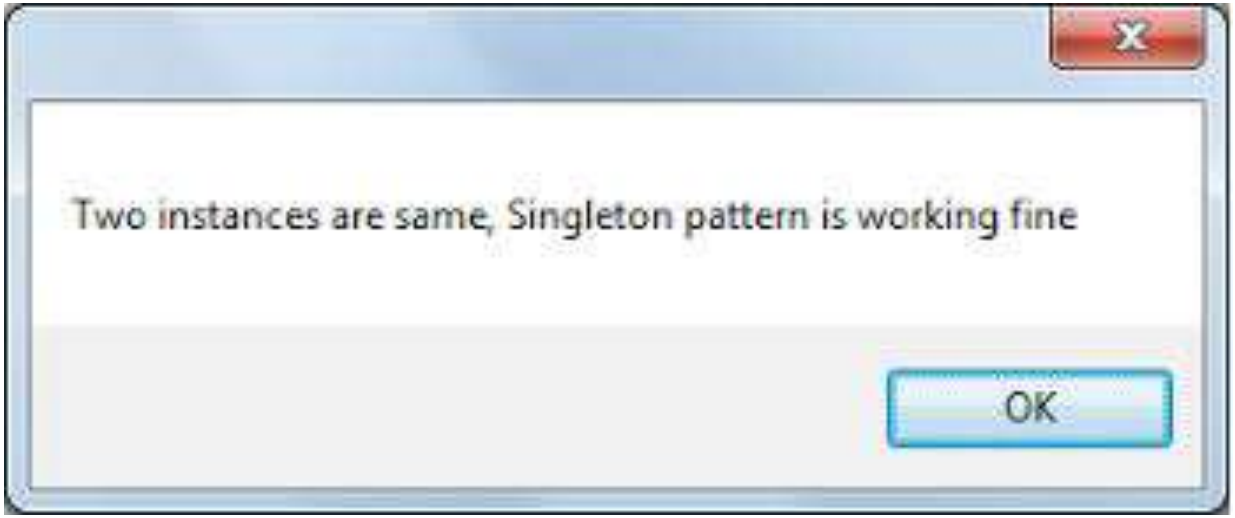


Prototype – The Good, The Bad …

- Copy objects without using the concrete class
- Skip repeated initializations
- Quickly create complex objects
- It is very hard to clone complex classes with many references

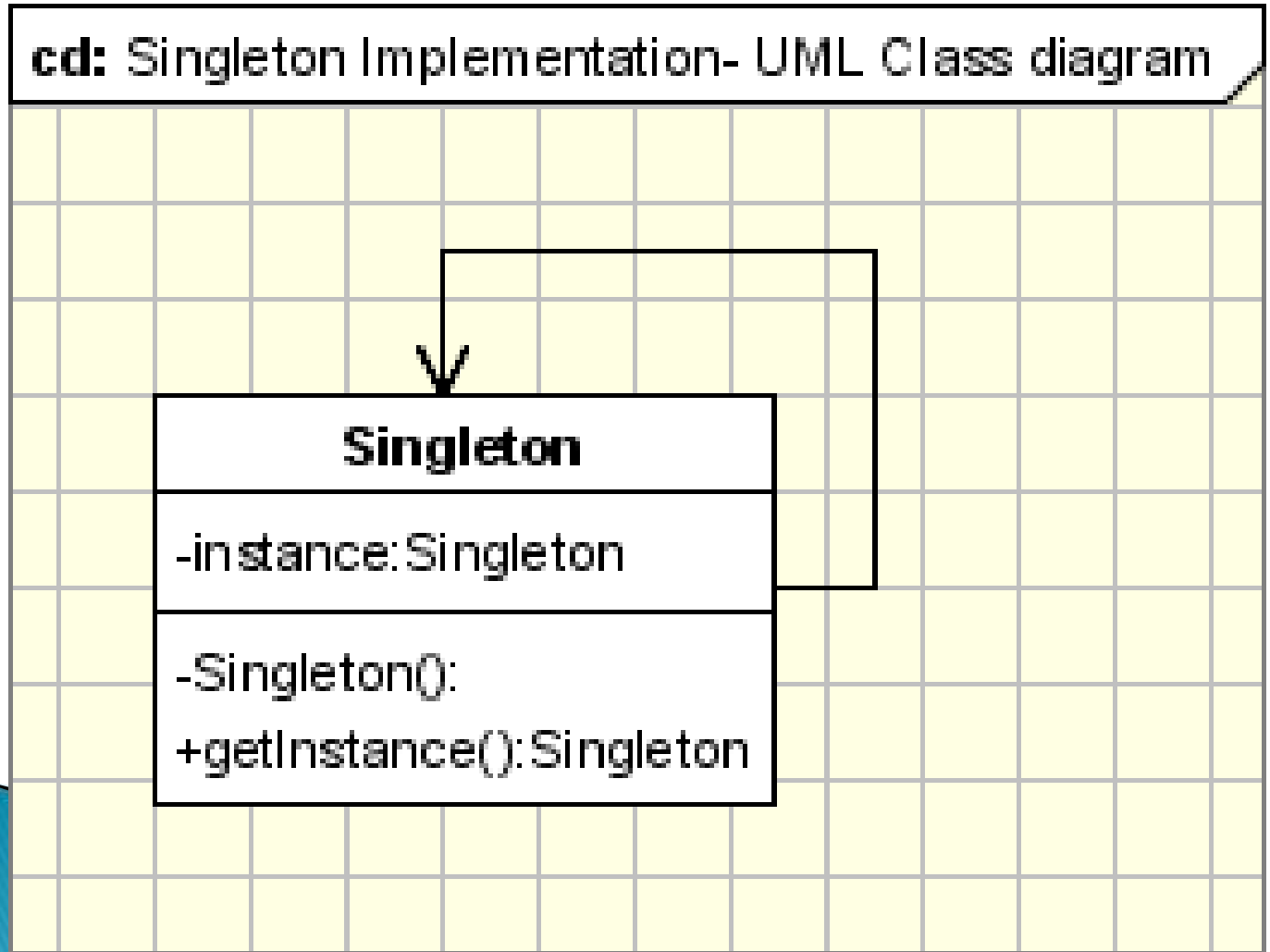
Creational Patterns – Singleton

- **Intent** – Ensure a class only has one instance, and provide a global point of access to it
- **Motivation** – It's important for some classes to have exactly one instance. There should be only one file system and one window manager. An accounting system will be dedicated to serving one company.
- How do we ensure that a class has only one instance and that the instance is easily accessible? A global variable makes an object accessible, but it doesn't keep you from instantiating multiple objects.


Two instances are same, Singleton pattern is working fine
OK

CP – Singleton

- **Singleton** – Logger



Singleton – The Good, The Bad …

- Guarantees a single instance
- Global access point to an instance
- Breaks S (from SOLID) – the pattern has TWO responsibilities
- Can mask high coupling or other bad design
- Is difficult to use in multi thread systems

CP – Lazy initialization

- **Lazy initialization** is the tactic of delaying the creation of an object, the calculation of a value, or some other expensive process until the first time it is needed

- How? With a **flag**
- Each time when the object is called, the **flag** is tested:
  - If it is **ready**, it is **returned**
  - If **not**, it is **initialized**

Lazy initialization – Java 1

```
public class Fruit {
  private static final Map<String,Fruit> types = new
   HashMap<String, Fruit>();
  private final String type;

  //using a private constructor to force use of the factory method
  private Fruit(String type) { this.type = type; }

  public static synchronized Fruit getFruit(String type) {
    if(!types.containsKey(type)) {
      types.put(type, new Fruit(type)); // Lazy initialization
    }
    return types.get(type);
  }
}
```

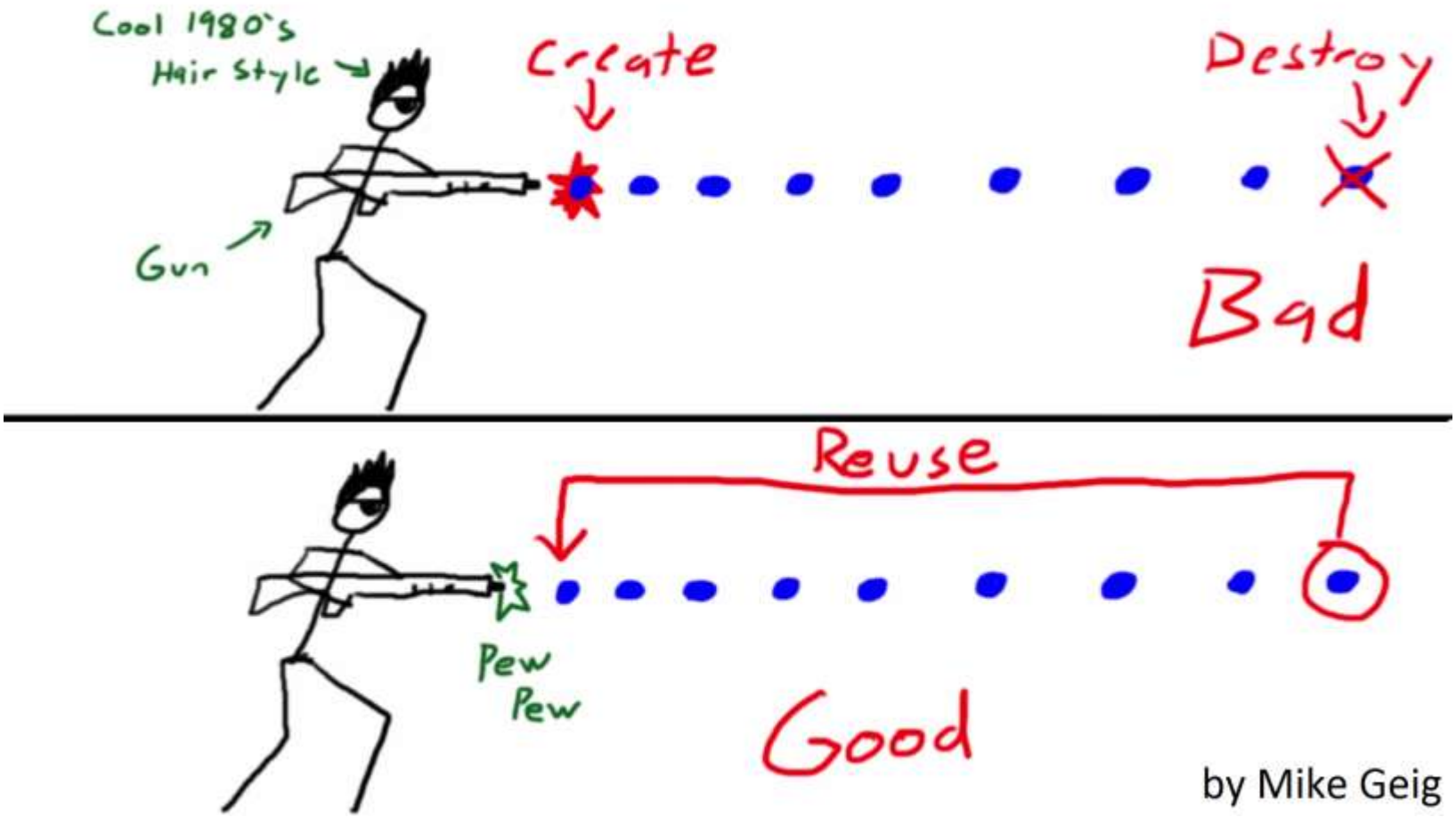
Lazy Initialization– The Good, The Bad …

- Saves on memory, since it instantiates only as needed
- More time consuming in case you need many new instances

CP – Object Pool

- **Intent**: reuse and share objects that are expensive to create.
- **Motivation**: Performance can be sometimes the key issue during the software development and **the object creation (class instantiation) is a costly step**. The Object Pool pattern offer a mechanism to reuse objects that are expensive to create
- **Why use it?** Basically, we'll use an object pool whenever there are several clients who needs the same stateless resource which is expensive to create

Visual Example of Object Pooling

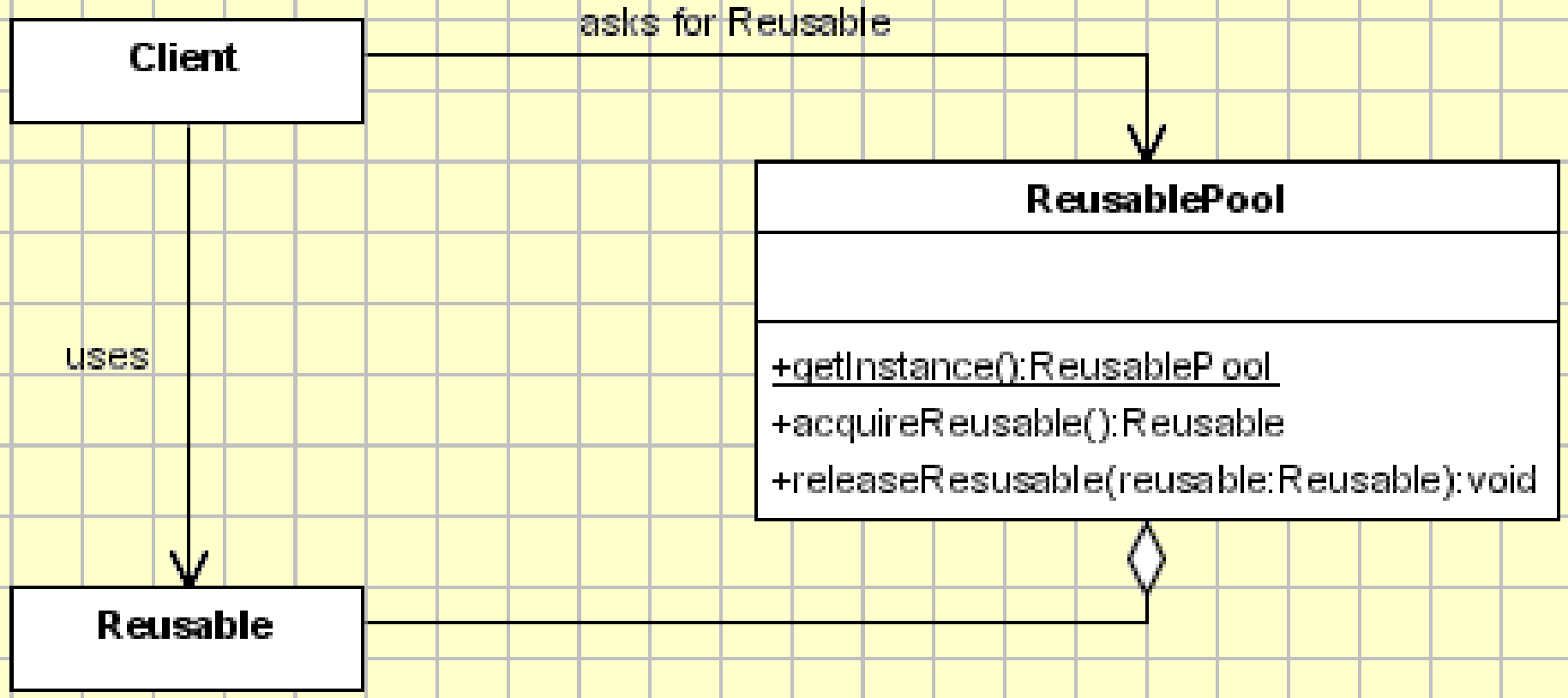


by Mike Geig

Object Pool – Implementation (1)

- When a client asks for a Reusable object, the pool performs the following actions:
  - **Search** for an available Reusable object and **if it was found** it will be **returned** to the client.
  - **If no Reusable object was found** then it **tries to create a new one**. If this actions **succeeds** the new Reusable object will be **returned** to the client.
  - If the pool **was unable to create a new Reusable**, the pool will **wait** until a reusable object will be released.

Object Pool – Implementation (2)



Object Pool – Applicability

- Lets' take the example of the **database connections**. It's obviously that opening too many connections might affect the performance:
  - **Creating a connection is an expensive** operation
  - When there are **too many connections opened** it **takes longer to create a new one** and the database server will become overloaded

- Here the **object pool comes to manage the connections** and provide a way to reuse and share them. It can also limit the maximum number of objects that can be created

Object Pool– The Good, The Bad …

- Saves on computing resources, since it only allows for a set number of instances
- Increased overhead in terms of managing the pool
- Slow for the objects in the waiting queue

Structural Patterns 1

- **Structural patterns are concerned with how classes and objects are composed to form larger structures**

- **Example**: consider how **multiple inheritance** mixes two or more classes into one. The result is a class that combines the properties of its parent classes. This pattern is particularly useful for making independently developed class libraries work together.

Structural Patterns 2

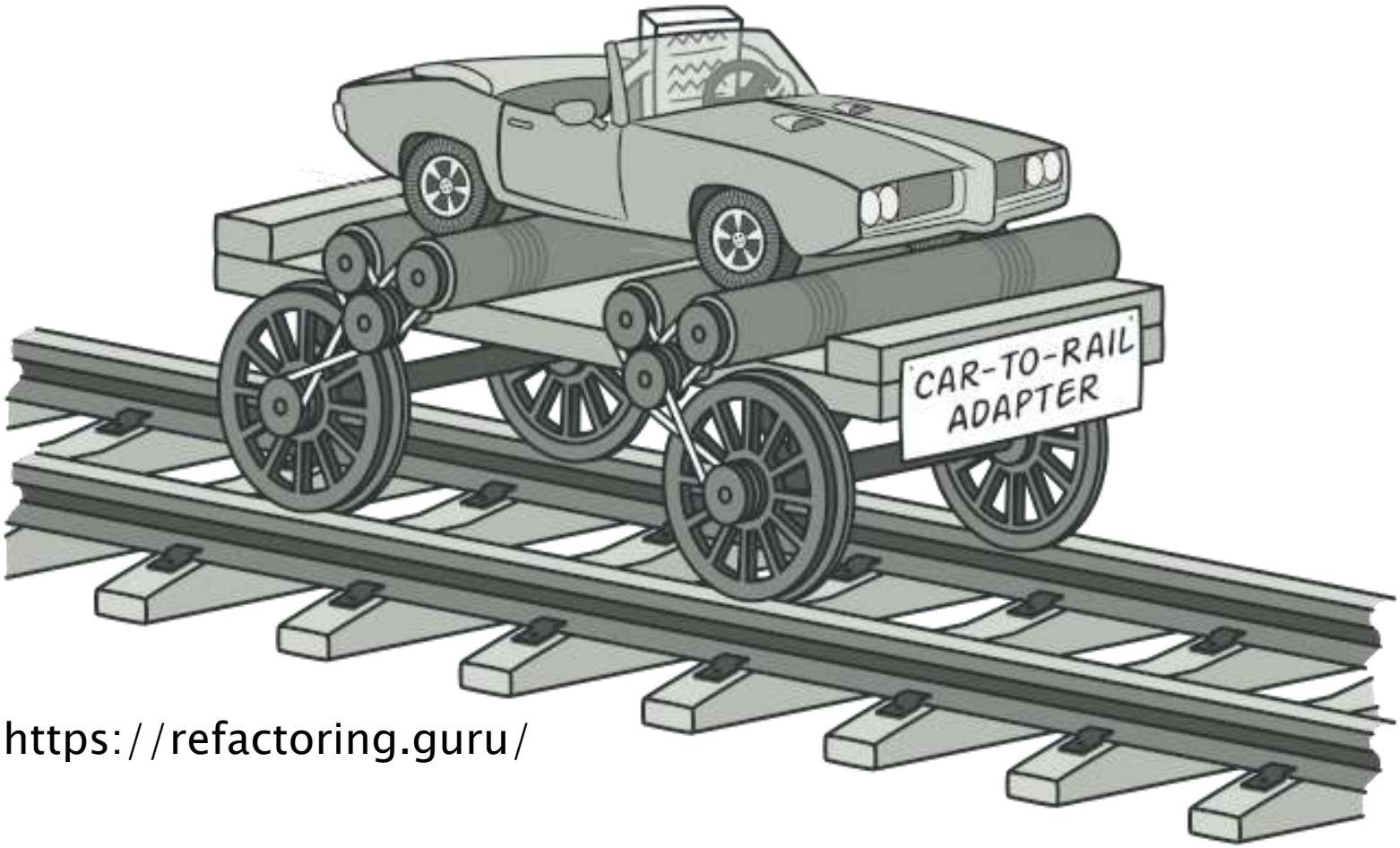
- Rather than composing interfaces or implementations, structural *object patterns* describe ways to compose objects **to realize new functionality**
- **The added flexibility of object composition** comes from the ability to change the composition at run-time

Structural Patterns

- Adapter
- Bridge
- Composite
- Decorator
- Façade
- Flyweight
- Proxy

Structural Patterns – Adapter

- **Intent**: Convert the interface of a class into another interface clients expect. Adapter lets classes work together that couldn't otherwise because of incompatible interfaces.
- **Also Known As**: Wrapper
- **Motivation**: Consider for example a drawing editor that lets users draw and arrange graphical elements (lines, polygons, text, etc.) into pictures and diagrams. The interface for graphical objects is defined by an abstract class called Shape. The editor defines a subclass of Shape for each kind of graphical object: a LineShape class for lines, a PolygonShape class for polygons, and so forth.



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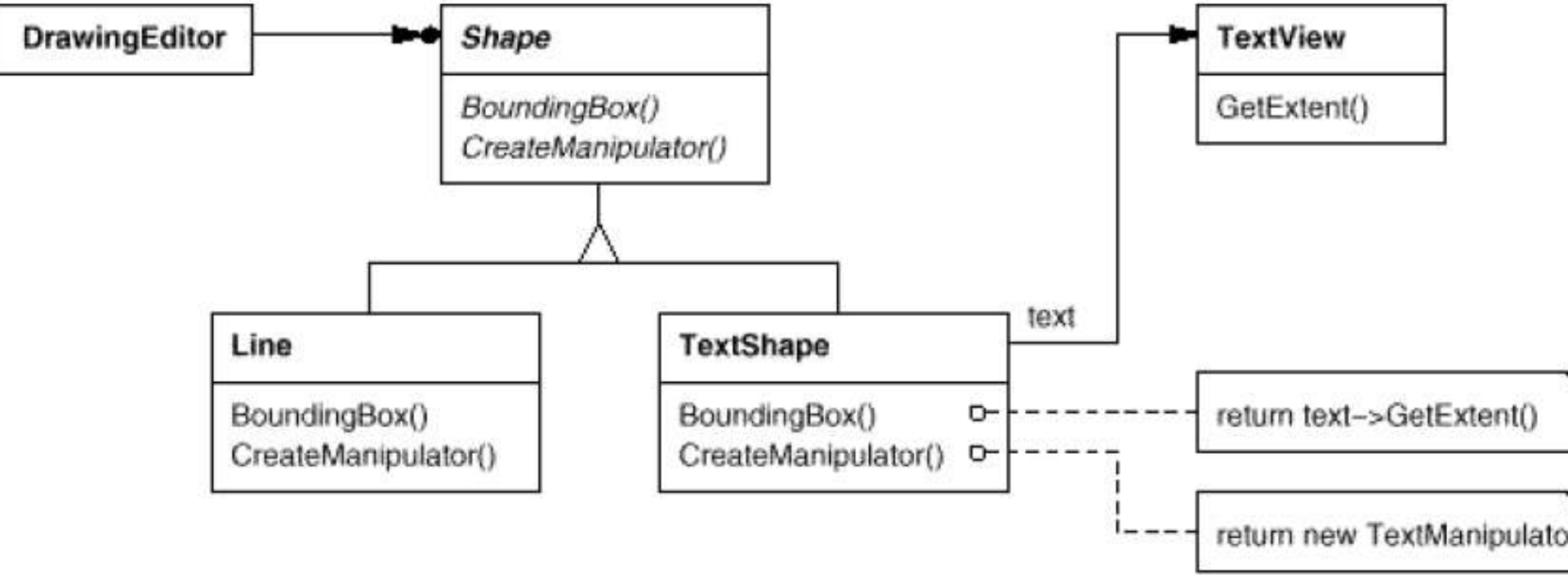
microSD
UHS-I 104MB/s
SD
UHS-I 104MB/s
USB-C
5Gb/s data speed
2 x USB 3.1 Gen 1
5Gb/s data speed
as well as high power
to your devices
HDMI Video Output
1080p @ 60Hz
Up to 4K @ 30Hz
Thunderbolt 3
Up to 5K or 2 X 4K @ 60Hz Video
Up to 40Gb/s data speed
Max 100W Power Delivery

Adapter 1

- A TextShape subclass that can display and edit text is considerably more difficult to implement
- We suppose the existence of a TextView class for displaying and editing text. So we can consider TextShape derived from these classes

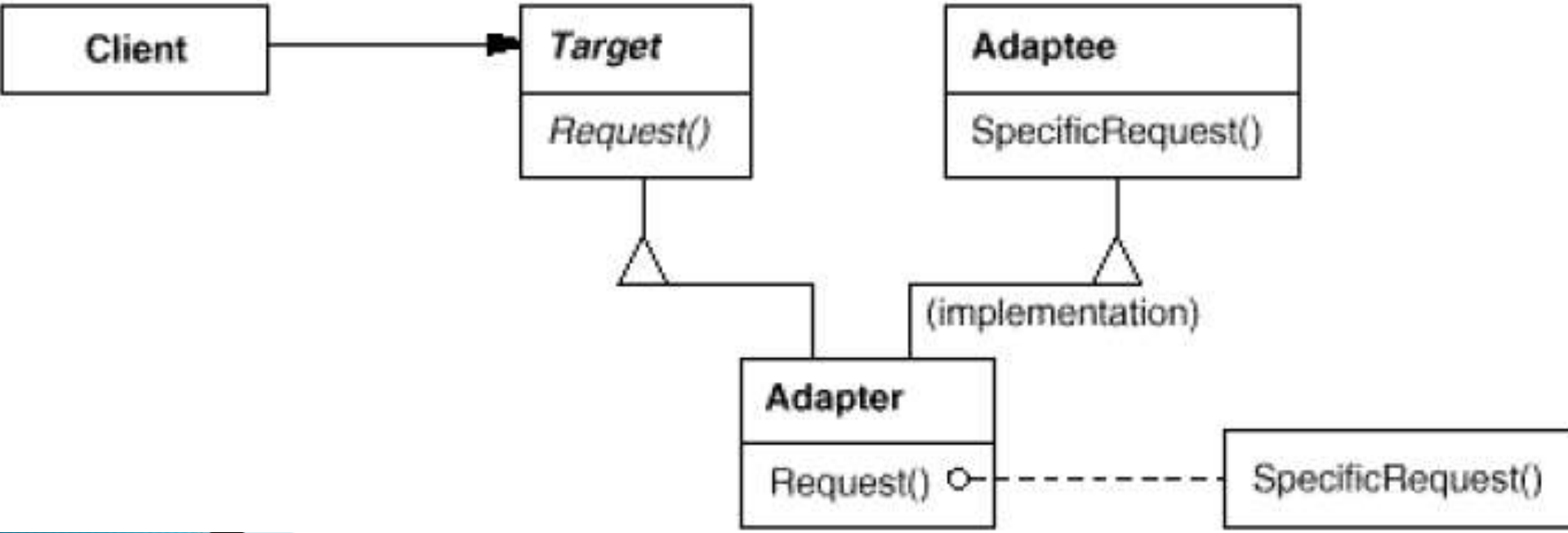


Adapter 2

- **Applicability** – Use the Adapter pattern when
  - you want to use an existing class, and its interface does not match the one you need
  - you want to create a reusable class that cooperates with unrelated or unforeseen classes, that is, classes that don't necessarily have compatible interfaces
  - *(object adapter only) you need to use several existing subclasses, but it's* impractical to adapt their interface by sub classing every one.

Adapter 3

- Structure – The Adapter pattern is used so that two unrelated interfaces can work together. This is something like we convert interface of one class into interface expected by the client.



Adapter – Example

- We all have electric sockets in our houses of different sizes and shapes. I will take an example of a socket of 15 Ampere. This is a bigger socket and the other one which is smaller is of 5 Ampere. A 15 Amp plug cannot fit into a 5 Amp socket => we will use an Adapter

- We have a 5 Amp plug and want a 5 Amp socket so that it can work. We DO NOT have a 5 Amp socket, what we have is a 15 Amp socket in which the 5 Amp plug cannot fit. The problem is how to cater to the client without changing the plug or socket.

Adapter – Java 1

```
/** The socket class has a specs for 15 AMP.*/
public interface Socket {
        public String getOutput();
}

public class Plug {
  private String specification = "5 AMP";
  public String getInput() {
        return specification;
  }
}
```

Adapter – Java 2

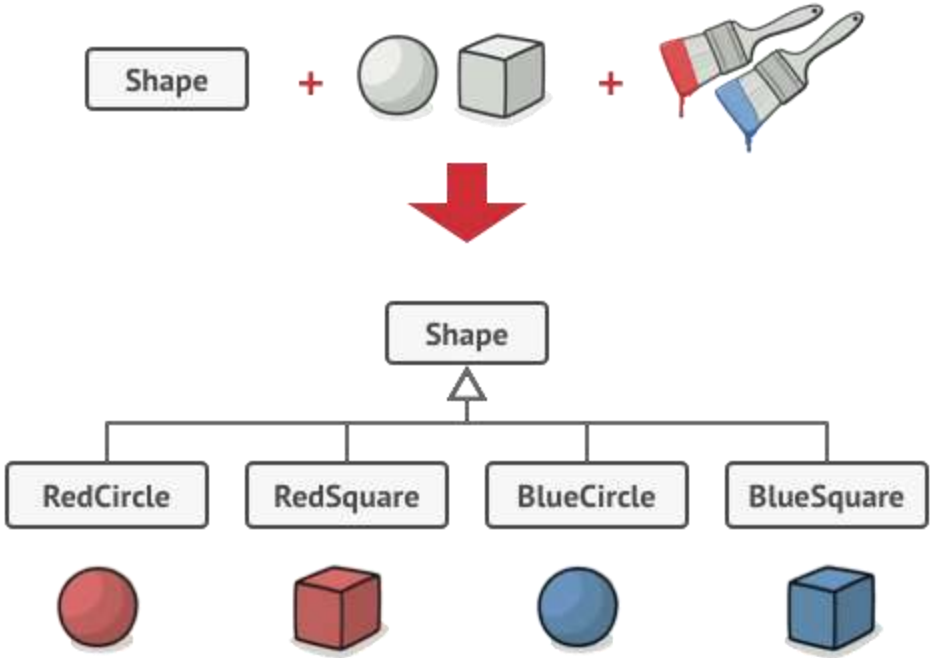
```
public class ConnectorAdapter implements Socket
{
  public String getOutput() {
      Plug plug = new Plug();
      String output = plug.getInput();
      return output;
  }
}
```

Adapter – The Good, The Bad …

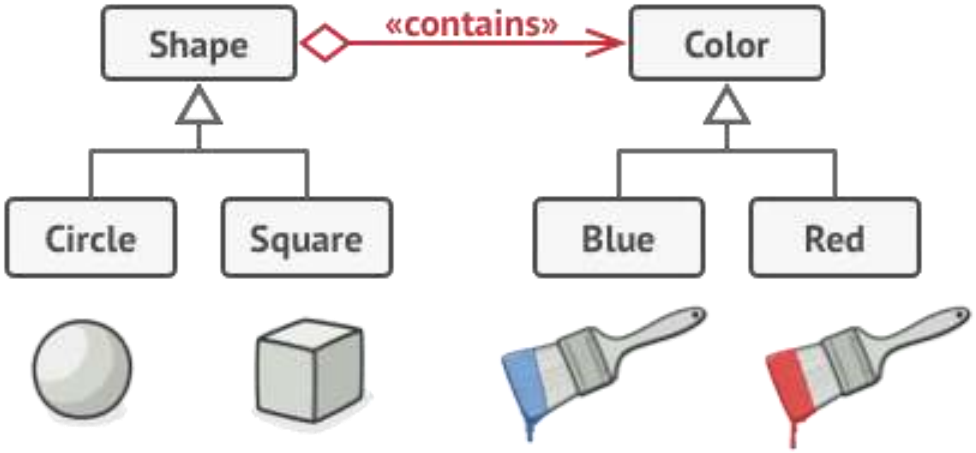
- Preserves S and O (from SOLID)
- Code becomes complicated because of extra classes

Structural Patterns – Bridge

- **Intent** – Decouple an abstraction from its implementation so that the two can vary independently
- **Also Known As** – Handle/Body
- **Motivation** – Consider the abstraction of shapes, each with its own properties. One thing all shapes can do is draw themselves. Drawing graphics to a screen can be dependent on different graphics implementations or operating systems. Shapes have to be able to be drawn on many types of operating systems



Vs.



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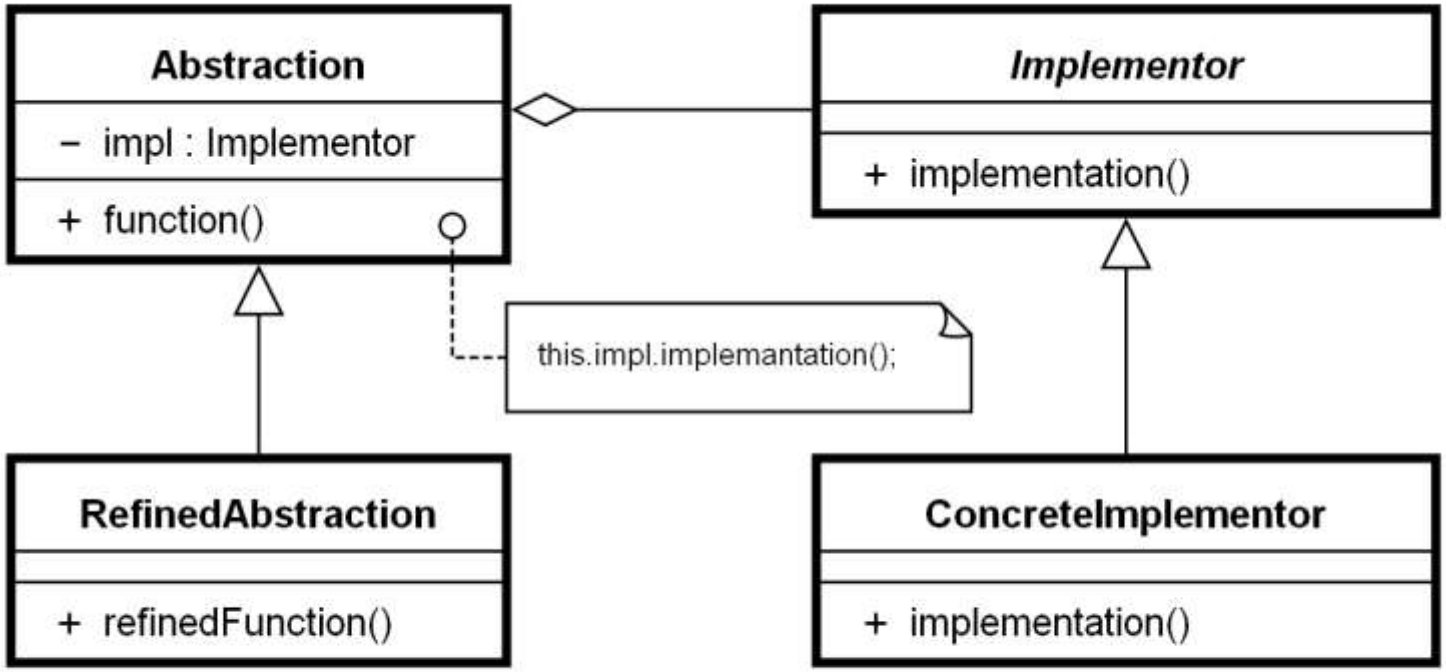
Bridge 1

- The bridge helps by allowing the creation of new implementation classes that provide the drawing implementation. Shape class provides methods for getting the size or properties of a shape. Drawing class provides an interface for drawing graphics



Bridge 2

- **Applicability** – When dividing large classes with many versions of the same feature (e.g. multiple types of network connections)
- When developing and extending classes in multiple and independent directions

Bridge – Java 1

```
interface DrawingAPI {/** "Implementor" */
  public void drawCircle(double x, double y, double
  radius);
}
/** "ConcreteImplementor" 1,2 */
class DrawingAPI1 implements DrawingAPI {
  public void drawCircle(double x, double y, double
  radius) { System.out.printf("API1” + x + y + radius);}
}
class DrawingAPI2 implements DrawingAPI {
  public void drawCircle(double x, double y, double
  radius) { System.out.printf("API2“ + x + y + radius);}
}
```

Bridge – Java 2

```
interface Shape {/** "Abstraction" */
   public void draw();
   public void resizeByPercentage(double pct); }

/** "Refined Abstraction" */
class CircleShape implements Shape {
   private double x, y, radius;
   private DrawingAPI drawingAPI;
   public CircleShape(double x, double y, double radius,
   DrawingAPI drawingAPI) {
         this.x = x; this.y = y; this.radius = radius;
         this.drawingAPI = drawingAPI;
   }
   public void draw() {drawingAPI.drawCircle(x, y, radius); }
   public void resizeByPercentage(double pct) { radius *= pct; }
}
```

Bridge – Java 3

```
/** "Client" */
class BridgePattern {
  public static void main(String[] args) {
      Shape[] shapes = new Shape[2];
      shapes[0] = new CircleShape(1, 2, 3, new DrawingAPI1());
      shapes[1] = new CircleShape(5, 7, 8, new DrawingAPI2());
      for (Shape shape : shapes) {
          shape.resizeByPercentage(2.5);
          shape.draw();
      }
  }
}
```

Bridge – The Good, The Bad …

- Preserves S and O (from SOLID)
- Allows for platform independent systems
- High level connections
- Code becomes complicated and splits cohesive classes

Bibliografie

- Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides: *Design Patterns: Elements of Reusable Object-Oriented Software* (GangOfFour)

Links

- Design Patterns: http://www.oodesign.com/
- Gang-Of-Four: http://c2.com/cgi/wiki?GangOfFour, http://www.uml.org.cn/c%2B%2B/pdf/DesignPatterns.pdf
- Design Patterns Book: http://c2.com/cgi/wiki?DesignPatternsBook
- About Design Patterns: http://www.javacamp.org/designPattern/
- Design Patterns – Java companion: http://www.patterndepot.com/put/8/JavaPatterns.htm
- Java Design patterns: http://www.allapplabs.com/java_design_patterns/java_design_patterns.htm
- Overview of Design Patterns: http://www.mindspring.com/~mgrand/pattern_synopses.htm
- Lazy initialization: http://en.wikipedia.org/wiki/Lazy_initialization
- Use Lazy Initialization to Conserve Resources: http://www.devx.com/tips/Tip/18007
- Python design patterns https://refactoring.guru/design-patterns